LEARNER GUIDE

Proof and problem-solving

Circle Theorems · Lesson 10. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

An angle chase finds values; a **proof** shows a result is *always* true. A proof is an **ordered chain of statements**: each one follows from the givens or from statements already made, and each carries a **reason**. The order is not free — you cannot use a fact before you have established it. This lesson builds that discipline. The great example is the keystone itself: **the angle at the centre is twice the angle at the circumference**. Draw P on the circle and extend PO to Q; the two radii make two isosceles triangles, the base angles give x and y, the exterior angles give $2x$ and $2y$ at the centre, and adding them gives $\angle AOB = 2(x+y) = 2\angle APB$. Once one theorem is proved it becomes a tool for the next — that is how all of mathematics is built.

What you are learning to notice

- A proof is an **ordered** chain — each step uses only what is already established.
- **Every statement has a reason**; a value with no reason is not a proof.
- A proved theorem becomes a **tool** for proving the next.
- A measurement checks one case; **only a proof** settles every case.

Moving through it, one step at a time

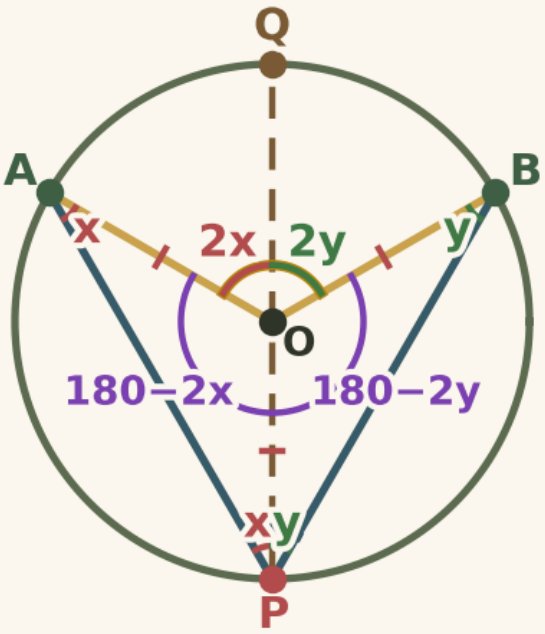
Start from the givens and a clear construction. Write each statement with its reason, in an order where every step already has what it needs. For a 'show that' problem, work towards the required statement, justifying each move, and finish by stating clearly that the target now follows. Keep asking: 'have I justified this, or only assumed it?'

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Prove the angle at the centre theorem

P is on the circle; A and B are two more points; O is the centre. Prove that ∠AOB = 2∠APB.

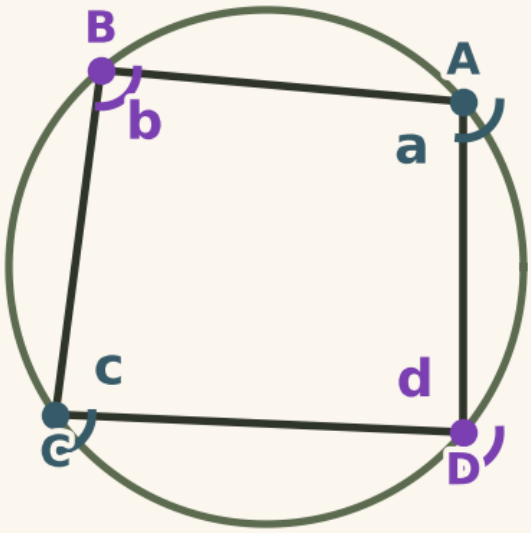


1. Join PO and extend it to Q. OA = OB = OP (radii), so triangles OAP and OBP are isosceles.	all radii are equal
2. ∠OAP = ∠OPA = x and ∠OBP = ∠OPB = y.	base angles of isosceles triangles
3. ∠AOQ = $2x$ and ∠BOQ = $2y$.	exterior angle = sum of the two opposite interior angles
4. ∠AOB = ∠AOQ + ∠BOQ = $2x + 2y = 2(x + y)$, and ∠APB = $x + y$.	adding the angles

∠AOB = 2∠APB — the angle at the centre is twice the angle at the circumference. ■

Example 2 — Prove the cyclic quadrilateral theorem

ABCD is a cyclic quadrilateral. Prove that ∠ABC + ∠ADC = 180°, using the theorem just proved.



1. Let ∠ABC = b and ∠ADC = d. Draw radii OA and OC.	set up the angles
2. ∠AOC = 2b (angle at the centre, standing on arc ADC with ∠ABC).	angle at the centre = 2 × angle at circumference
3. The reflex angle AOC = 2d (standing on arc ABC with ∠ADC).	angle at the centre, again
4. 2b + 2d = 360° (angles around O), so b + d = 180°.	angles around a point

∠ABC + ∠ADC = 180° — opposite angles of a cyclic quadrilateral are supplementary. ■

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (all the circle theorems, isosceles base angles, the exterior-angle fact, and that angles around a point add to 360°), open a short recap and return whenever you are ready.

- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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